

SimVille: Modulating Conversational Demand in an LLM-Driven VR Social Game via Emotional Tone and Multiparty Interaction

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Figure 1: Game scene with a conversation five virtual characters talking about ordering food.

ABSTRACT

Everyday social conversations can be challenging, particularly in situations involving social pressure or conflict. We investigate whether immersive virtual reality (VR) can provide a safe and controllable environment in which such conversations can be systematically designed to be challenging, thereby offering a “playground” for users to practice and prepare for demanding real-life interactions. In our game *SimVille*, players complete scenarios through spoken interaction with embodied NPCs in a situated virtual environment, where dialogue is dynamically generated using Large Language Models (LLMs). We empirically examine how manipulating conversational demand (emotional tone and dyadic vs. multiparty interaction) affects perceived workload and affective experience.

In a controlled user study ($N = 25$), negative-emotion NPC dialogue (negative emotional tone) significantly increased perceived mental workload, temporal demand, effort, and frustration/stress compared to positive/neutral dialogue (happy tone), while the multiparty (five NPCs) compared to the dyadic (one NPC) scenario increased mental demand and effort. Overall, our findings demonstrate how adjustable workload can support personalized and scal-

able conversation practice.

Index Terms: Large Language Model, Virtual Reality, Games for Health and Wellbeing, Emotional Everyday Scenarios

1 INTRODUCTION

Social conversations in everyday life can become demanding, particularly in situations involving social pressure, disagreement, or interpersonal conflict. These challenges are especially relevant for people who experience difficulties in social communication, as prior work shows that well-designed immersive environments can improve social understanding and emotion regulation when tailored to cognitive and sensory needs [20, 14].

The presented work is intended as a formative pre-study for future applications with autism spectrum disorder (ASD) users in Virtual Reality (VR). The system design followed a user-centered process informed by input from autistic people and psychologists; however, the reported evaluation provides first insights based primarily on a non-autistic participant sample to assess feasibility, controllability, and user experience before targeted deployment. In contrast to therapist-led sessions, immersive VR roleplay scenarios can be delivered in a scalable and accessible manner, supporting more frequent practice outside of clinical settings. A key advantage of VR is that immersive environments can reduce unwanted external stimuli, allowing users to focus on conversations without sensory overload and directing attention to the virtual interaction context [34]. In addition, VR enables users to rehearse socially difficult conversations in a safe and controlled environment, which supports gradual exposure to challenging situations [23]. Unlike desktop-based chat systems, immersive VR supports spatialized audio and

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world-anchored object manipulation, enabling realistic social complexity (e.g., background conversations, group interactions, and interruptions) while maintaining experimental control.

Within this context, *SimVille* investigates how naturalistic, emotionally expressive virtual agents can support conversational learning in a socially rich VR setting. The system integrates immersive VR with adaptive LLM-driven dialogue to simulate cooperative and challenging social encounters. The emotional tone plays a central role in shaping the cognitive and affective demands of social interaction. Previous research shows that negatively valenced or conflicting emotional signals can increase conversational load, increase stress responses, and require greater perspective-taking during human-agent interaction [25]. From a cognitive point of view, users must allocate additional resources for emotion recognition, regulation, and Theory of Mind (ToM) reasoning when interlocutors display impatience, sarcasm, or frustration.

The study investigates how the number of conversational partners and their emotional tone influence user engagement, comfort, and cognitive load in a virtual restaurant scenario. Two research questions guide the experiment:

1. **RQ1:** How does the emotional tone of NPCs influence users' emotional experience and perceived cognitive load?
2. **RQ2:** How does the conversational setting (dyadic vs. multiparty) affect the user experience across social interaction scenarios?

Our contributions are:

- A **controllable framework** for VR social conversation that operationalizes along two dimensions: (1) **emotional tone** (positive vs. negative) and (2) **scenario type** (dyadic vs. multiparty interaction).
- Evidence that conversational manipulation is reflected not only in self-report, but also in **observable linguistic behavior**, including user/NPC sentiment as estimated via transcript-based sentiment analysis.
- Design implications for building **adaptive, personalized VR conversational training systems**, showing how emotional tone and group size can function to support progressive exposure and skill development.

2 RELATED WORK

This section reviews related work on VR-based social interaction and games for health and wellbeing.

2.1 Virtual Reality for Controlled Social Interaction

Previous work has demonstrated that immersive VR environments can support social skills training by enabling users to engage in realistic, repeatable interactions with virtual characters under controlled conditions [15]. Immersive VR environments can be used to systematically investigate the dynamics and presence of social interactions by comparing different interaction conditions and environmental variables [9]. In particular, the results show that the presence and behavior of virtual crowds can significantly influence users' perceived social pressure, emotional state, and sense of presence, underscoring the impact of social density on user experience. XR-based interventions [8] exist in multiple developmental domains, including social interaction, recognition and regulation of emotions, communication, attention, and reduction of anxiety. A literature review on immersive technologies for teaching social skills to students with autism spectrum disorder highlights the potential of VR-based interventions to provide controlled, repeatable, and engaging social learning experiences that can improve social communication outcomes [33]. A virtual reality-based framework

was developed to assess emotional everyday scenarios with varying levels of conversation partners as a foundation for future digital health applications targeting autistic people. The study combined subjective ratings and psychophysiological measures (ECG, EDA) to validate that immersive environments can differentiate emotional and safety responses across conditions, demonstrating the therapeutic potential of VR for social and emotional training [45].

Artificially embodied AI agents in immersive virtual reality can exert measurable social influence on human users, shaping their perceptions, decisions, and behaviors at levels comparable to human-controlled counterparts [28]. Similarly, a 2×2 design within-subjects was used to systematically examine how different configurations of a generative AI assistant and varying levels of task complexity affect user performance and cognitive load in immersive VR learning environments [10]. More broadly, surveys on social interaction in VR highlight the importance of agent behavior, emotional expression, and social realism as key determinants of presence and engagement [25].

Virtual Reality Exposure Therapy (VRET) has been widely studied as an intervention for anxiety-related conditions such as public speaking anxiety and social anxiety disorder. Prior work suggests that VRET can achieve efficacy comparable to traditional in vivo exposure while offering advantages in accessibility, controllability, and potentially lower drop-out rates. However, challenges related to immersion and presence remain, and further research is needed on self-guided VRET, safety, and ethical aspects such as data protection and quality control [39].

Another work investigates how different interaction modalities with LLM-driven non-player characters in VR, such as speech-based interaction versus traditional dialog boxes, influence users' sense of presence and immersion [11]. The results indicate that while speech-based interaction can enhance immersion, it may also increase cognitive load, and conversational latency remains a key challenge for maintaining seamless social engagement in virtual environments.

Personality traits of LLM-controlled embodied conversational agents in VR significantly influence users' social evaluations, emotional experience, and engagement during face-to-face interactions with virtual partners [24]. These findings highlight how immersive VR with adaptive agent behavior can shape socio-emotional processing and interactive behavior in controlled virtual social environments.

In social VR contexts, ELLMA-T, an embodied LLM agent designed for English language learning in VRChat—demonstrates the potential of LLMs to enable realistic, context-aware, and adaptively scaffolded role-play interactions [35]. Artificial social influence through embodied conversational agents based on LLM to build relationships for health coaching demonstrates how adaptive, empathetic dialog strategies can strengthen user trust, participation, and persuasive impact in behavior change interventions [29]. Environment-aware, LLM-driven agents for virtual reality role-play scenarios illustrate how integrating spatial context and object-level awareness into dialogue systems enables more coherent, context-sensitive, and immersive interactions [27]. A comprehensive review of LLM-driven approaches for VR digital humans, highlighting challenges such as latency, computational constraints, and the lack of standardized evaluation frameworks [7].

Previous research has demonstrated that VR can serve as an effective medium to improve social skills in people with ASD [20]. However, the intensity and complexity of sensory stimuli within VR environments necessitate careful adaptation to the individual's perceptual and cognitive needs [14]. Consequently, it is essential to design VR experiences that account for varying levels of difficulty, ensuring that each participant remains within an optimal range of motivation and engagement without experiencing excessive stress or sensory overload.

The use of virtual avatars in assistive learning has gained increasing attention in recent research. Accurate representations of virtual humans are particularly important for training social skills in immersive environments, as they allow users to develop a deeper sense of empathy, especially when the virtual character is able to provide empathic feedback [5]. Systematic reviews highlight substantial benefits for learners, including improvements in social skills, emotional competencies, daily living abilities, communication, and attention through interactions with avatars [21, 31].

2.2 Large Language Models for Social and Interactive Agents

Ongoing research explores the capabilities of LLMs in diverse domains, including arithmetic, common sense reasoning, and symbolic processing [18], as well as linguistics, understanding of the physical world, temporal reasoning, and social intelligence [4].

LLMs also show strong potential to mimic aspects of daily human life: evaluation studies demonstrate that generative agents [36] can produce individual and emergent social behaviors through structured cycles of planning, execution/scoring and reflection. These components enable agents to recall prior interactions and proactively initiate contextually appropriate actions in response to user input. In gaming contexts, LLMs have demonstrated the ability to simulate human life by reproducing interactions among individuals within a virtual village [36]. Beyond entertainment, AI agents, chatbots, and virtual assistants are increasingly being applied in healthcare, where they can provide support and disseminate information using specialized training data [17].

Gamified online role-play has also shown promise; for example, a teledentistry study reported improved self-perceived confidence and awareness when a professional actress was trained to play a patient with a dental condition [42]. Furthermore, experimental studies demonstrate that the integration of empathic modules - capable of simulating emotional states and offering clarifications - significantly enhances perceived empathic listening and emotional intelligence, thus strengthening learner trust [1]. Unlike scripted agents, generative agents introduce conversational unpredictability and emergent social behavior, which can increase mental model demand and attentional demands [36, 1]. Previous work suggests that participants may even prefer to interact with virtual avatars over to interact with a human interventionist in certain contexts [32].

Furthermore, LLMs have demonstrated the ability to simulate reasoning of the mental state, achieving performance comparable to humans in ToM assessments [41, 40, 3, 37].

3 METHOD

This section explains the framework of the test environment and the methods applied to answer our research questions.

3.1 Scenario Design

SimVille is designed as an immersive conversational training game located in everyday social contexts, primarily within a restaurant setting. Players complete two scenarios that systematically vary social and emotional workload by manipulating both the conversational setting (dyadic vs. multiparty interaction) and the NPCs’ emotional tone (positive vs. negative). This structure targets conversational competencies such as small talk, turn-taking, and emotion regulation.

After each interaction, a supportive companion NPC provides verbal feedback generated by the LLM, reinforcing effective behaviors (e.g., maintaining calmness, initiating conversation). The *Conversational Circumplex* [46] highlights how conversational motives and constraints shape human dialogue, which aligns with *SimVille* scenarios in which players must navigate social and logical obstacles (e.g., misunderstandings or conflicting conversational tones) to

achieve successful interactions. Our study design results in four experimental conditions.

A 2×2 factorial structure was implemented, as summarized in Tab. 1, with the following independent variables:

- **Emotional Tone of NPCs:** All NPCs expressed either a *positive* or *negative* affective style. Emotional tone was informed by Ekman’s taxonomy of basic emotions [13]. Positive conditions featured friendly behaviors aligned with emotions such as happiness or surprise, whereas negative conditions employed unfriendly behaviors linked to anger, disgust, fear, or sadness, producing clearly differentiated emotional climates within the virtual environment.
- **Scenario Type (Dyadic vs. Multiparty):** Participants either engaged in a *dyadic scenario* with a single NPC (one-on-one interaction; low social coordination demands) or in a *multiparty scenario* with a group of five NPCs (group interaction; increased social attention and turn-taking demands), as shown in Figs. 2 and 3.



Figure 2: Example scene with one conversation partner on the desk in the burger restaurant.



Figure 3: Example scene with five conversation partners on the desk in the burger restaurant.

To simulate realistic social interactions and challenge the player’s conversational reasoning, each scenario in *SimVille* introduces a subtle logical or social twist. These twists require participants to adapt flexibly, drawing on social understanding as well as problem-solving abilities. In addition to spoken dialog, each scenario incorporates a task-based interaction with the virtual environment. For example, during purchase scenarios, players obtain virtual food items, as shown in Fig. 4, and deliver them to specific

Scenario	Task	LLM Emotion Instruction
Dyadic Scenario (Positive Emotion)	(A) The user orders food in a simple one-on-one interaction with a single NPC. The user confirms Barbara’s cheeseburger + cola order and brings it back.	(A) Barbara behaves friendly, happy, and supportive; replies are short, warm, and cooperative.
Dyadic Scenario (Negative Emotion)	(B) The user orders food in a one-on-one interaction while Barbara is stressed and in a meeting. The interaction remains simple but emotionally tense.	(B) Barbara behaves angry, rushed, and impatient; replies are short, irritated, and tense.
Multiparty Scenario (Positive Emotion)	(C) The user interacts with a group of five work colleagues inside Big Burger. The group must reach a shared lunch decision (Classic Menu vs. Fitness Menu). Social coordination demands are high, but emotional tone is friendly.	(C) NPCs behave warm, supportive, and cooperative; emotional states are positive (e.g., happy, surprised, neutral).
Multiparty Scenario (Negative Emotion)	(D) The user interacts with five work colleagues with conflicting preferences. The group must reach (or fail to reach) a shared lunch decision. Social coordination demands are high and emotional tone is challenging.	(D) NPCs behave grumpy, angry, disgusted, fearful, or sad; interpersonal conflicts occur (e.g., NPCs disagree with each other).

Table 1: Matrix of scenario types (dyadic vs. multiparty), and LLM emotional instructions. The dyadic scenario consists of one NPC, whereas the multiparty scenario consists of five NPCs.

NPCs, thus coupling conversational decision-making with embodied manipulation of objects in the world. This design leverages VR to ground communication in situated action and to support immersive, hands-on interaction beyond purely verbal exchange.

Dyadic Scenario - Ordering Food In the dyadic scenario, the player interacts with a single NPC colleague at the restaurant counter. The task is to order a cheeseburger and a drink for the colleague Barbara. The scenario includes a logical obstacle: cola is unavailable on the menu, requiring the player to negotiate alternatives (e.g., ordering Pepsi instead). Additionally, a budget constraint (total cost ≤ 6 euros) can only be met by recognizing a discounted combo option. This scenario evaluates the player’s ability to maintain goal-oriented conversation, handle incomplete information, and remain polite under varying emotional tone. After purchasing the food, the user can use the VR controllers to carry and deliver the ordered items to Barbara.

Multiparty Scenario - Group Lunch Decision In the multiparty scenario, the player interacts with a group of five colleagues seated at a table. The task is to guide the group toward a shared decision between two lunch options (Classic Menu vs. Fitness Menu). This setting increases conversational demand by requiring the player to distribute attention across multiple interlocutors, interpret competing social cues, and manage turn-taking. In a friendly environment, the colleagues express mild preferences, but remain cooperative and open to compromise. In the unfriendly condition, emotional tension and interpersonal conflict arise, with some NPCs rejecting certain options or refusing to agree with others. This scenario provides a controlled test of social coordination and emotion regulation under multiparty conversational dynamics. NPCs also engage in dialogue with each other, requiring the user to continuously monitor and follow the ongoing group conversation. After purchasing the food, the user can use the VR controllers to carry and deliver the ordered items to the group.

3.2 Framework

SimVille enables naturalistic spoken interaction through a speech-based interface, where players communicate with NPCs via a push-to-talk mechanism. This interaction design prevents unintended speech recognition activations while preserving low-friction verbal dialogue suited to in-situ VR conversations. Beyond direct NPC communication, the environment incorporates spatialized ambient audio (e.g. background chatter) that becomes increasingly audible as players approach groups of NPCs engaged in conversation. This contextual soundscape enhances social plausibility and immersion by conveying ongoing social activity outside the focal interaction,



Figure 4: Navigating *SimVille* while transporting purchased food items to deliver them to designated NPCs.

thereby reinforcing the perception of a populated and socially active virtual environment. The overall system architecture is shown in Fig. 5.

The prototype system was developed in Unity 6000.2.2f1 using the XR Interaction Toolkit to create immersive virtual reality environments and manage user interactions in the Meta Quest 3 headset. The application ran on a workstation equipped with an NVIDIA RTX 4070 Ti GPU, an Intel Core i7-14700K processor and 32 GB of RAM. Avatars were implemented using Ready Player Me characters, which were imported into Unity with integrated rigging, facial blendshapes, and synchronized lip movement. Virtual characters were animated using assets from Mixamo, with gestures and animations to maintain coherence between verbal and non-verbal communication.

For natural language understanding and dialogue generation, the system employed the Gemma 3 12B model, accessed locally through the LLM for Unity plugin. We configured the LLM with a low temperature ($T = 0.1$) and a context window of 16,384 tokens to promote stable and consistent responses. This approach controls LLM variability and ensures that responses remain consistent across participants. The model was selected to ensure that the game environment, the speech-to-text system, and the LLM could operate efficiently within the 16 GB of available GPU memory.

This integration enabled real-time natural language exchange between users and NPCs. Text-to-speech (TTS) and speech-to-text (STT) were implemented using OpenAI models, with gpt-4o-mini-tts providing expressive voice output that enhanced social presence and conversational realism. The used language for all conversations is English.

Our repository for the game and all log data is available on

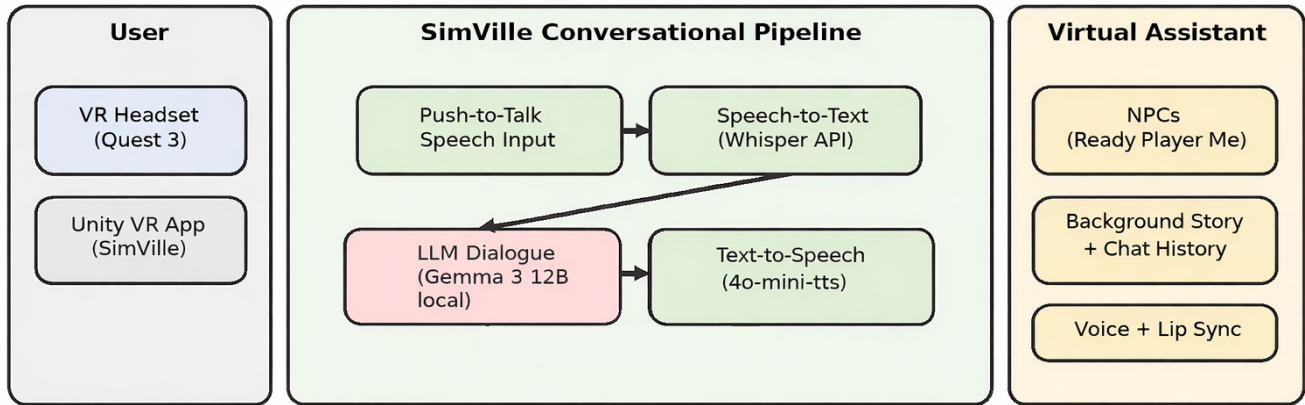


Figure 5: System architecture of SimVille’s conversational pipeline, including speech input, STT, local LLM dialogue generation, and TTS output for NPC interaction.

github¹.

3.3 Prompt Design

Prompts defined the conversational behavior of NPCs and evaluation agents within each condition. Each role was described in natural language to guide the LLM’s dialogue generation while maintaining experimental control over tone and engagement.

Supportive NPC Defines a cooperative agent that facilitates calm, encouraging interaction.

LLM Instruction for Supportive NPC

You are Bea, a friendly designer at SimVille. You recently moved to the city from a small coastal town, bringing a bright artistic energy that quickly made you popular in the office. You are in a happy, loving relationship with Donald; the two of you often exchange playful glances during meetings, and Mario occasionally jokes that you’re the “team power couple.” You’re close with Sami, appreciating his calm nature, which balances your bubbly side. You and the team are discussing which lunch option to choose inside Big Burger. You prefer the Fitness Menu. Your emotional tone is positive (happy). You speak in 1–3 warm, gentle, cooperative sentences.

Conflict-Driven NPC. Defines an agent that introduces mild social tension and emotional conflict.

LLM Instruction for Conflict-Driven NPC

You are Bea, a designer at *SimVille* who is currently angry especially toward Donald. Your relationship with him ended badly, leaving you irritable and confrontational whenever he speaks. Your anger also makes you stubborn: you insist on the Fitness Menu, and the idea of agreeing with Donald or backing down only fuels your frustration. You and the team are discussing which lunch option to choose, and the tension is palpable. Your emotional tone is angry. You speak in 1–3 sharp, tense sentences.

¹blinded

Scenario evaluation Defines a role responsible for providing adaptive performance feedback and updating the game state based on conversational outcomes (e.g., confirming payment and spawning the purchased food item on the counter so the user can deliver it to another NPC).

LLM Instruction for Evaluation Agent

Based on the chat history evaluate the question:
Did the user pay?
After evaluation respond with exactly one character: 1 if Yes, 0 if No. If the outcome is unclear, respond with 0.

3.4 Study Design

The study was carried out in full compliance with ethical guidelines and standards. All procedures were reviewed and approved by the Ethics Committee of blinded.

A power analysis conducted using G*Power 3.1 for repeated-measures ANOVA (medium effect size $f = .25$, $\alpha = .05$, power $1 - \beta = .80$) indicated a minimum required sample size of 24 participants.

All participants player all four scenarios in a randomized order.

3.5 Measures

In this study, our inferential analyses focus on subjective workload and affective experience measures. In addition, we complement these self-report outcomes with sentiment analysis of spoken user utterances as a language-based indicator of how conversational workload shapes participants’ verbal responses.

- **Pre-study questionnaire.** Participants completed a short questionnaire assessing demographic information (e.g., age and sex) and self-reported ASD diagnosis status.
- **In-game speech recordings and transcripts.** During each scenario, participants’ spoken utterances were recorded and transcribed. Subsequently, these transcripts were analyzed using VADER sentiment scores to derive an automated language-based indicator of the expressed affective tone.
- **Post-scenario questionnaires.** After each scenario, participants rated their experience using:

- **Subjective workload (NASA-TLX).** The NASA Task Load Index (NASA-TLX; [16]) is a multidimensional instrument that assesses subjective workload across six dimensions: mental, physical, and temporal demand, perceived performance, effort, and frustration. Items are rated on a 7-point Likert scale (1–7), and the mean across subscales yields an overall workload score.
- **Perceived emotional tone.** Participants answered: “What kind of emotion did this scene evoke in you?” using six basic emotions [13] (happiness, sadness, anger, fear, surprise, disgust) plus neutral.
- **Post-study questionnaire (presence and comfort).** At the end of the session, participants completed a questionnaire assessing comfort and presence in VR using the Igroup Presence Questionnaire (IPQ-Short). The IPQ [38, 2, 44, 30] measures the subjective sense of “being there” in virtual environments and comprises three subscales *Spatial Presence*, *Involvement*, and *Realness*, along with a general presence item.

To automatically quantify the affective valence of conversational utterances, we applied the Valence Aware Dictionary and sEntiment Reasoner (VADER). Positive and neutral expressions typically yield non-negative compound scores, whereas negatively valenced expressions result in lower or negative scores [19].

4 RESULTS

All participants experienced all four experimental conditions. A total of $N = 25$ participants completed the survey ($n = 14$ male, $n = 11$ female). Participants had a mean age of $M = 32.68$ years ($SD = 6.17$; range: 23–43). Two participants self-reported an ASD diagnosis ($n=2$), while the remaining participants did not ($n=23$).

4.1 Task Load

Analysis of the individual NASA-TLX dimensions revealed distinct effects across workload components and directly addressed our research questions regarding the influence of *emotional tone* (RQ1) and *dyadic vs. multiparty scenario* (RQ2) on perceived workload.

As shown in Fig. 6, perceived workload increased for emotionally challenging (unfriendly) interactions and, for some workload components, also for scenarios involving five NPCs. The strongest differences were observed for mental demand, temporal demand, effort, and frustration, consistent with the repeated-measures ANOVA (RM-ANOVA) results.

Mental demand showed significant main effects of both factors. Participants reported higher mental demand when interacting with five NPCs compared to one NPC (RQ2), $F(1, 24) = 14.34$, $p = .0009$, $\eta_p^2 = .374$, as well as higher mental demand in the challenging compared to the positive condition (RQ1), $F(1, 24) = 7.79$, $p = .010$, $\eta_p^2 = .245$. The interaction did not reach significance, $F(1, 24) = 3.09$, $p = .092$, $\eta_p^2 = .114$.

For *temporal demand*, we found a strong main effect of *emotional tone* (RQ1), indicating increased time pressure in the challenging condition, $F(1, 24) = 15.06$, $p = .0007$, $\eta_p^2 = .386$. No main effect of *dyadic vs. multiparty scenario* was observed, $F(1, 24) = 0.04$, $p = .848$, $\eta_p^2 = .002$. However, the interaction between *dyadic vs. multiparty scenario* and *emotional tone* was significant, $F(1, 24) = 4.36$, $p = .0477$, $\eta_p^2 = .154$, suggesting that the effect of emotional tone on perceived time pressure differed between single- and multiparty conversations.

Effort ratings were significantly influenced by both experimental factors. Participants reported higher effort when interacting with five NPCs compared to one NPC (RQ2), $F(1, 24) = 16.51$, $p = .0004$, $\eta_p^2 = .408$, and higher effort under challenging compared to positive conditions (RQ1), $F(1, 24) = 13.95$, $p = .0010$,

$\eta_p^2 = .368$. The interaction effect was not significant, $F(1, 24) = 1.04$, $p = .319$, $\eta_p^2 = .041$, indicating predominantly additive effects of partner count and emotional tone on perceived effort.

Subjective *performance* ratings showed a significant main effect of *emotional tone* (RQ1), with lower performance in the challenging compared to the positive condition, $F(1, 24) = 5.20$, $p = .0317$, $\eta_p^2 = .178$. The main effect of *dyadic vs. multiparty scenario* did not reach significance, $F(1, 24) = 3.84$, $p = .0619$, $\eta_p^2 = .138$, and no interaction was observed, $F(1, 24) = 0.00$, $p = .950$, $\eta_p^2 = .000$.

For *physical demand*, no significant main effects of *dyadic vs. multiparty scenario* ($p = .095$) or *emotional tone* ($p = .096$) were observed. However, we found a significant interaction, $F(1, 24) = 7.30$, $p = .0125$, $\eta_p^2 = .233$, indicating that the effect of emotional tone on physical demand differed depending on whether participants interacted with one versus five NPCs.

Finally, *frustration/stress* showed a strong main effect of *emotional tone* (RQ1), indicating increased emotional strain in the challenging condition, $F(1, 24) = 9.79$, $p = .0046$, $\eta_p^2 = .290$. Neither the main effect of *dyadic vs. multiparty scenario* (RQ2), $F(1, 24) = 2.61$, $p = .1195$, $\eta_p^2 = .098$, nor the interaction reached significance, $F(1, 24) = 1.61$, $p = .2167$, $\eta_p^2 = .063$.

Overall, the NASA-TLX results provide clear support for both research questions. With respect to **RQ2** (dyadic vs. multiparty scenario), interacting with five NPCs significantly increased *mental demand* and *effort*, and showed a trend towards reduced perceived *performance*. Regarding **RQ1** (emotional tone), the challenging condition significantly increased *mental demand*, *temporal demand*, *effort*, and *frustration/stress*, while significantly decreasing subjective *performance*. Two significant interaction effects (*temporal demand* and *physical demand*) indicate that different emotions can affect workload differently in dyadic compared to multiparty conversations.

4.2 Sentiment Analysis

We analyzed affective tone using VADER compound sentiment scores computed for each utterance. For inferential testing, compound scores were aggregated per participant and condition (i.e., mean compound sentiment per *Scenario Type* $\times$ *Emotional Tone* cell). Two-way repeated-measures analyses of variance (RM-ANOVA) were conducted separately for *User* and *NPC* utterances, with *Scenario Type* (dyadic vs. multiparty) and *Emotional Tone* (Positive, Negative) as within-subject factors. Tab. 3 summarizes the RM-ANOVA results for compound sentiment scores.

Descriptive Sentiment Scores. Table Tab. 4 reports mean VADER compound scores for User and NPC utterances across all scenarios. Overall, NPC utterances exhibited higher positive sentiment than user utterances, with the largest differences observed in the positive condition.

User Sentiment. For user utterances, the RM-ANOVA revealed a significant main effect of **scenario type** on compound sentiment, $F(1, 24) = 9.43$, $p = .006$. In addition, there was a significant main effect of the **emotional tone**, $F(1, 24) = 15.25$, $p < .001$, indicating that user sentiment differed between the positive and negative conditions. No significant interaction between scenario and emotional tone was observed, $F(1, 24) = 0.00$, $p = .967$.

NPC Sentiment. For NPC utterances, the RM-ANOVA revealed significant main effects of **Scenario Type** (dyadic vs. multiparty), $F(1, 24) = 35.38$, $p < .001$, and **Emotional Tone** (positive vs. negative), $F(1, 24) = 408.91$, $p < .001$. Importantly, a significant **Scenario Type** $\times$ **Emotional Tone** interaction was observed, $F(1, 24) = 6.89$, $p = .016$, indicating that the effect of emotional tone on NPC sentiment differed between dyadic and multiparty scenarios.

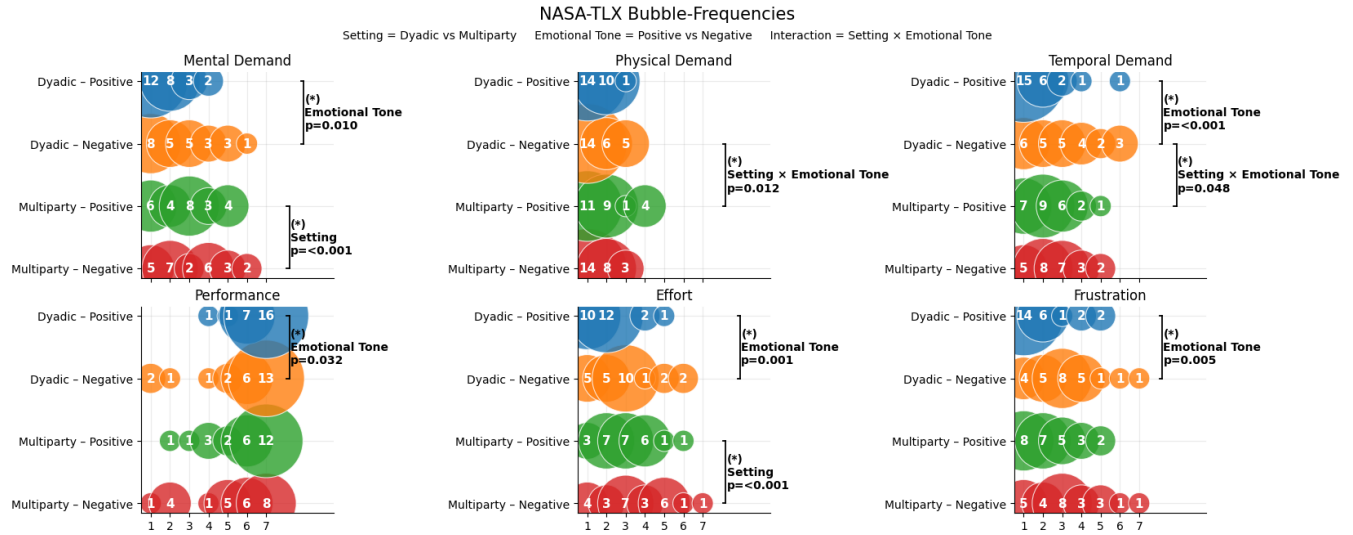


Figure 6: NASA-TLX subscale ratings across experimental conditions (Setting × Emotional Tone). Circle size represents response frequency. Significant effects ($p < .05$) are annotated.

Table 2: Results of 2×2 repeated-measures ANOVAs (RM-ANOVA) for NASA-TLX subscales ($N = 25$). Significant effects ($p < .05$) are shown in bold.

Measure	Scenario Type (Dyadic vs. Multiparty)	Emotional Tone (Positive vs. Negative)	Scenario Type × Emotional Tone
Mental Demand	$F(1, 24) = 14.34, p < .001, \eta_p^2 = .374$	$F(1, 24) = 7.79, p = .010, \eta_p^2 = .245$	$F(1, 24) = 3.09, p = .092, \eta_p^2 = .114$
Physical Demand	$F(1, 24) = 3.02, p = .095, \eta_p^2 = .112$	$F(1, 24) = 3.00, p = .096, \eta_p^2 = .111$	$F(1, 24) = 7.30, p = .013, \eta_p^2 = .233$
Temporal Demand	$F(1, 24) = 0.04, p = .848, \eta_p^2 = .002$	$F(1, 24) = 15.06, p < .001, \eta_p^2 = .386$	$F(1, 24) = 4.36, p = .048, \eta_p^2 = .154$
Performance	$F(1, 24) = 3.84, p = .062, \eta_p^2 = .138$	$F(1, 24) = 5.20, p = .032, \eta_p^2 = .178$	$F(1, 24) = 0.00, p = .950, \eta_p^2 < .001$
Effort	$F(1, 24) = 16.51, p < .001, \eta_p^2 = .408$	$F(1, 24) = 13.95, p = .001, \eta_p^2 = .368$	$F(1, 24) = 1.04, p = .319, \eta_p^2 = .041$
Frustration / Stress	$F(1, 24) = 2.61, p = .120, \eta_p^2 = .098$	$F(1, 24) = 9.79, p = .005, \eta_p^2 = .290$	$F(1, 24) = 1.61, p = .217, \eta_p^2 = .063$

Table 3: Two-way RM-ANOVA results for VADER compound sentiment scores. Effects are reported for *scenario type* and *emotional tone*. Significant effects ($p < .05$) are shown in bold.

Speaker	Effect	F	p
User	Scenario Type	$F(1, 24) = 9.43$	.006
User	Emotional Tone	$F(1, 24) = 15.25$	< .001
User	Scenario Type × Emotional Tone	$F(1, 24) = 0.00$	.967
NPC	Scenario Type	$F(1, 24) = 35.38$	< .001
NPC	Emotional Tone	$F(1, 24) = 408.91$	< .001
NPC	Scenario Type × Emotional Tone	$F(1, 24) = 6.89$	.016

Table 4: Mean VADER compound sentiment scores (with utterance counts in parentheses).

Scenario Type	Emotional Tone	User	NPC
Dyadic	Negative	0.127 (221)	0.241 (227)
Dyadic	Positive	0.200 (207)	0.642 (208)
Multiparty	Negative	0.165 (281)	0.370 (366)
Multiparty	Positive	0.229 (240)	0.746 (324)

4.3 Emotions

The distribution of self-reported emotions differed across the experimental conditions, see Tab. 5. Conditions with a positive emotional tone primarily evoked positive or neutral emotions, with *Happy* responses most frequent in the dyadic positive condition. In contrast, conditions with a negative emotional tone elicited more negative emotional responses, most notably *Angry* in the dyadic negative condition. Overall, negative-tone interactions were associated with

a broader distribution of negatively valenced emotions compared to positive-tone interactions.

Because each participant reported an emotion in all four conditions (within-subject design), we did not apply a chi-square test of independence. Instead, we conducted paired McNemar tests by binarizing each emotion (selected vs. not selected) and comparing conditions within participants. Holm correction was applied within each contrast to control the family-wise error rate across emotions.

For the dyadic setting, the negative condition significantly reduced *Happy* responses compared to the positive condition (McNemar, Holm-corrected $p = .0036$; $n_{10} = 15$ switched from *Happy* to not-*Happy* vs. $n_{01} = 1$ switched from not-*Happy* to *Happy*). Conversely, *Angry* responses significantly increased from dyadic positive to dyadic negative (McNemar, Holm-corrected $p = .0234$; $n_{01} = 9$ switched to *Angry* vs. $n_{10} = 0$ switched away from *Angry*). No further emotion contrasts reached significance after Holm correction.

Table 5: Absolute counts of detected emotions per scenario condition.

Scenario / Condition	Happy	Surprised	Neutral	Sad	Disgust	Angry
Dyadic - Positive	15	4	5	0	0	0
Dyadic - Negative	0	5	6	1	3	9
Multiparty - Positive	8	4	11	0	1	0
Multiparty - Negative	4	4	9	0	0	6

Table 6: Post-questionnaire results (IPQ-Short; 1–7 Likert; $N = 25$). Higher values indicate greater presence/immersion.

Item	M	SD
Inside virtual environment	4.92	1.55
Deeply engaged	4.60	1.58
Environment appeared realistic	3.60	1.73
Feeling of being part of environment	5.20	1.50
Distracted by real world (raw)	1.60	0.65
Inside virtual environment (question)	5.04	1.31
Not distracted (reverse-coded)	6.40	0.65
IPQ-Short overall (mean; distraction reversed)	4.96	1.12

4.4 Immersion

Participants reported moderate-to-high presence in *SimVille* (IPQ-Short overall: $M = 4.96$, $SD = 1.12$), suggesting that the VR implementation effectively supported a sustained sense of “being there” during the interactive scenarios. Ratings indicated a strong sense of being inside the environment (e.g., “inside the VE”: $M = 4.92$, “being part of VE”: $M = 5.20$) and low distraction from the real world (reverse-coded: $M = 6.40$), consistent with the high level of attentional capture typically enabled by immersive head-mounted VR and embodied first-person interaction. Perceived realism was comparatively lower ($M = 3.60$), indicating that while immersion and spatial presence were successfully elicited, the visual and/or social fidelity of the virtual setting (e.g., environment or agent behavior) may limit subjective realism. Perceived realism was comparatively lower, likely due to the use of low-poly environmental assets and stylized avatars, which may have reduced visual fidelity despite maintaining a strong sense of presence.

5 DISCUSSION

In this section, we discuss our findings and the limitations of our study. Overall, our results indicate that conversations in immersive VR can be tuned along two largely independent dimensions: emotional tone and social interaction type (dyadic vs. multiparty scenario type). Across NASA-TLX subscales (Tab. 2, Fig. 6), emotionally challenging LLM-driven NPC dialogue significantly increased perceived workload, most prominently reflected in higher temporal demand, effort, and frustration/stress, as well as reduced perceived performance. These findings suggest that negative-emotion interactions can function as an effective “affective stressor” in VR-based conversational practice.

In addition, we found that the multiparty conversational setting (five NPCs) increased cognitive load relative to the dyadic setting (one NPC), particularly increasing mental demand and effort (Tab. 2). This aligns with the increased need to allocate attention across multiple interlocutors, manage turn-taking, and respond to competing social cues in spatially situated conversations.

Prior XR and conversational agent research provides strong motivation for our research questions on **emotional tone** (RQ1) and **dyadic vs. multiparty interaction** (RQ2). Regarding RQ2, studies in immersive VR suggest that conversational coordination demands can raise cognitive workload; for example, long multimodal conversations in VR highlight a trade-off between richer interaction and increased cognitive load [43], and work on collaborative social VR further indicates that task structure and role demands modulate NASA-TLX workload dimensions [22]. Regarding RQ1, prior work on conversational agents shows that negative or problematic agent behavior can increase user frustration and adverse affect [6]. In addition, multiparty conversational settings introduce coordination challenges such as addressee selection and turn-taking management [12], suggesting that emotional tone may differentially affect user experience in dyadic versus multiparty scenarios.

5.1 Findings

Our results suggest that conversational challenge in immersive role-play can be intentionally shaped along multiple, partially independent dimensions. In particular, modulating NPC *emotional tone* provides an “affective stressor” that increases subjective workload and frustration, while introducing *multiparty* interactions increases social-cognitive demands associated with attention management and turn-taking. Practically, designers can construct individualized training ladders by varying tone and group size to support gradual exposure, while monitoring user workload and affect to prevent overwhelming experiences.

RQ1 (Emotional Tone). The observed effects of emotional tone on perceived workload align with prior research on affective demands in immersive virtual environments, where emotionally challenging interactions increase cognitive and temporal workload by requiring additional resources for emotion recognition, regulation, and perspective-taking [16, 26, 13]. This pattern was reflected in our results: the negative-emotional NPC dialogue (angry/sad tones) significantly increased overall perceived workload, as well as temporal demand, effort and frustration, while physical demand remained largely unaffected. Furthermore, the use of NPCs driven by LLM may have amplified these effects, since generative agents introduce conversational variability, unpredictability, and emerging social behaviors that increase the complexity of the mental model and the attentional demands compared to scripted dialogue [36, 1].

RQ2 (Dyadic vs. multiparty scenario). Prior work in immersive VR has shown that interacting with multiple embodied agents can alter user experience and increase cognitive load and decision pressure compared to single-agent interactions [28, 10]. Consistent with this, our findings demonstrate that the multiparty scenario (five NPCs) compared to the dyadic scenario (one NPC) significantly increased mental demand and effort. Beyond this main effect, our results suggest that conversational demands do not scale linearly with the number of interlocutors: the significant *interaction of the emotional tone type $\times$ emotional tone* in the sentiment of NPC indicates that the negative-emotion dialog can disproportionately shape the affective climate of multiparty conversations. This may increase the need to manage turn-taking, competing social cues, and shifting emotional dynamics [46].

5.2 Limitations

Although *SimVille* enabled dynamic spoken conversations with LLM-driven NPCs, the interaction between user behavior and the game world remained limited. User input primarily influenced dialogue content, while the surrounding environment and narrative progression were only minimally reactive (e.g., limited consequences of conversational choices). Future work should integrate deeper world-state modeling, enabling meaningful cause–effect relationships between user actions, NPC behaviour, and scenario outcomes (e.g., branching scenarios, persistent social relationships, and task success/failure states). Such a tighter coupling between user input and the virtual environment could increase ecological validity and further support personalized training.

We did not specifically target or recruit autistic individuals; clinical validation remains future work. Although *SimVille* is developed within a larger project focusing on conversational support for autistic people, the present work represents an initial feasibility study evaluating the experience of controlled conversations in VR.

6 CONCLUSION

In this paper, we introduce *SimVille*, an LLM-driven VR social training game that enables users to practice conversational skills through natural spoken interaction with embodied NPCs in everyday social scenarios. In a controlled study 2×2 within-subjects ($N = 25$), we showed that negative emotional NPC dialogue (e.g. angry/sad tones) significantly increased perceived workload and

frustration, while multiparty (five NPCs) vs dyadic (one NPC) scenario increased mental demand and effort. Together, these results demonstrate that emotional tone and dyadic vs multiparty scenario are powerful design parameters for adjustable experience, supporting scalable personalization of VR-based conversational training and wellbeing interventions.

SUPPLEMENTAL MATERIALS

All supplemental materials and the codebase is available on github at [blinded](#).

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